

# Arin Shahbazian

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📍 Blacksburg, Virginia

🌐 [Webpage](#)

## Personal Statement

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As a PhD in Economics with expertise in health economics, causal inference, and quasi-experimental methods, I am committed to advancing evidence-based health and population policy through empirical research. My dissertation, collaborative research, and internship experience apply statistical and causal inference techniques to large-scale claims, EHR, survey, and RCT data to evaluate treatment effects, healthcare disparities, utilization, costs, and long-term health outcomes.

## Education

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- Ph.D. in Economics, Virginia Tech (VT) Jan 2021- Sep. 2026
- Master of Economics, Virginia Tech ([VT](#)) Jan 2021- Dec. 2022
- Master of Socio-Economic Systems Engineering, [SUT](#) Sep. 2013- Jan. 2016
- Bachelor in Applied Mathematics, [KNTU](#) Sep. 2007- Sep. 2012

## Recent Research Experience and Working Papers

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- **Arin Shahbazian**, Wen You, Xu Lin, Delia West, Becca Krukowski, “Time to Weight Loss and Subsequent Weight Maintenance: A Survival Analysis” (Job Market Paper)
- **Arin Shahbazian**, Wen You, Xu Lin, Delia West, Becca Krukowski, “The Role of Motivations in Weight Outcomes: An RCT Study”
- **Arin Shahbazian**, Wen You, Xu Lin, Delia West, Becca Krukowski, “Incentive Design and Long-Term Weight Management Success: A Counterfactual Policy Analysis”
- **Arin Shahbazian**, Asal Pilehvari, Becca Krukowski, Caroline Mulligan, “Machine Learning-Driven Insights into the Structural and Socioeconomic Predictors of Top Five Cancer Mortality Rates in the U.S.” (Under review at “Cancer Medicine” journal)
- Asal Pilehvari, **Arin Shahbazian**, Qingyi He, Gloribel Bonilla, Roger Anderson, “Impact of Treatment Delays on Healthcare Costs in Patients with Breast Cancer: A Mediation Analysis of Healthcare Utilization by Socioeconomic Status” (submitted to CMS for initial approval).
- Asal Pilehvari, **Arin Shahbazian**, Gloribel Bonilla, Roger Anderson, “Disparities in Treatment Delay and its Trends” (submitted to CMS for initial approval).
- Asal Pilehvari, **Arin Shahbazian**, “Machine Learning for Breast Cancer Stage Prediction in Claims Data: A SEER-Medicare Study” (in progress).
- Mackenzi Oswald, **Arin Shahbazian**, Kathryn M. Ross, Wen You, Becca Krukowski, “An Exploratory Analysis of Patterns of Weight Change in a Post-Cessation Weight Management Trial” (Published at Nicotine and Tobacco [Research](#), May 2025).

## Work Experience

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- **Data Science Graduate Intern** at [UAA, VIRGINIA TECH](#), VA, USA. May 2026 - Aug. 2026  
- Developed the Value Index, a composite scoring framework measuring alumni career, civic, and resilience outcomes, contributing to measure design and the analysis plan.
- **Research Intern** at [PUBLIC HEALTH DEP., UNIVERSITY OF VIRGINIA](#), VA, USA. Jan. 2025 - May 2025  
- Conducted healthcare claims research using SEER-Medicare data, applying causal inference and predictive methods to examine cancer treatment delays, healthcare utilization and costs, and disparities by socioeconomic status across several projects.
- **Data Analyst & Econometrician** at [NETBINA](#) Feb. 2019 - Jan. 2021  
- Analyzed Household Expenditures and Income Survey data to produce multi-scale analytical reports on consumer and market trends for brand partners.
- **Data Analyst & Researcher** at [PARLIAMENT RESEARCH CENTER](#) Mar. 2017 - Jan. 2021  
- Analyzed Household Expenditures and Income Survey data for poverty and health studies, producing technical reports for policymakers.
- **Researcher** at [INSURANCE RESEARCH CENTER \(IRC\)](#) Mar. 2016 - Oct. 2018  
- Conducted independent research on premium placement in household budgets and premium elasticity.
- **Economist** at [TEHRAN CHAMBER OF COMMERCE \(TCCIM\)](#) Jan. 2017 - Mar. 2017  
- Prepared economic reports for company managers.

## Key Skills

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- **Real-World Data:** Administrative claims (SEER-Medicare), clinical trial/RCT data, registry data, EHR
- **Models and Methods:** Instrumental variables (IV/2SLS), DiD, regression discontinuity, marginal structural models, propensity score matching/weighting, synthetic control, GMM, G-estimation, principal stratification, mediation analysis, panel data methods, Causal forests, predictive/statistical learning methods, survival analysis, event study, cost-effectiveness analysis, program and policy evaluation, Power analysis/MDE, resampling methods
- **Software:** R, Python, Stata, SAS (proficient)  
SQL, Julia, MATLAB, JMP, SPSS, Tableau, Power BI, Git/GitHub (working knowledge)
- **AI Tools:** Claude, ChatGPT, Gemini, Microsoft Copilot

## Conference Presentations

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- Time to Weight Loss and Subsequent Weight Maintenance: A Survival Analysis  
- Presented at Economics Graduate Student Conference ([EGSC](#)) at WashU, October 2024  
- Presented at Global ([GLO](#)) Conference, December 2025
- The Role of Motivations in Weight Outcomes: An RCT Study  
- Presented at 14th Annual Conference of the American Society of Health Economists ([ASHEcon](#)), June 2025
- Machine Learning-Driven Insights into the Structural and Socioeconomic Predictors of Top Five Cancer Mortality Rates in the U.S.  
- Presented at Annual Public Health Association ([APHA 2025](#)), November 2025

## Teaching Experience

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### Lecturer / Instructor of Record

INTERMEDIATE MICROECONOMICS, VT  
PRINCIPAL MICROECONOMICS, VT

(Teaching evaluation details are available [here](#) )

Spring 2026, Fall 2025, Fall 2024, Fall 2023, Summer 2023  
Winter 2026, Summer 2024

## Research Interests

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| - Health Economics | - Health Policy & Evaluation | - Health Disparities |
| - Causal Inference | - Applied Economics          | - Machine Learning   |

## Recent Awards and Honors

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- **Outstanding Graduate Instructor of the year**, [Department of Economics](#) , VT, 2024.
- **Armenian General Benevolent Union Scholarship (AGBU)**  Recipient, 2024-2025.
- **Armenian Student's Association Scholarship (ASA)**  Recipient, 2023-2025.
- **Full-Time Graduate Assistantship**, Department of Economics, VT (2021- present).
- **Graduate School Travel Fund Program Grant**, VT (Spring 2024, Spring 2025, Fall 2025).

## SERVICE ACTIVITIES

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- Vice President, Treasurer, and Secretary of Armenian Student Association at Virginia Tech (2023-2026)

## Language Proficiency

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ARMENIAN: Native

PERSIAN: Native

ENGLISH: Fluent

## Reference

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- **Dr. Xu Lin** (Ph.D. Dissertation Advisor)  
Associate Professor of Economics, Department of Economics, Virginia Tech  
Email: [xulin88@vt.edu](mailto:xulin88@vt.edu)
- **Dr. Djavad Salehi-Isfahani** (Faculty Supervisor in Research)  
Professor of Economics, Department of Economics, Virginia Tech  
Email: [salehi@vt.edu](mailto:salehi@vt.edu)
- **Dr. Asal Pilehvari** (Internship's Supervisor)  
Assistant Professor of Public Health Sciences, Department of Public Health, University of Virginia  
Technical Director, Population Health and Cancer Outcomes Core, UVA Cancer Center  
Email: [asal@virginia.edu](mailto:asal@virginia.edu)
- **Dr. Wen You** (Committee Member)  
Professor of Public Health Sciences, Department of Public Health, University of Virginia  
Director, Population Health and Cancer Outcomes Core, UVA Cancer Center  
Adjunct Professor, Department of Agricultural & Applied Economics, Virginia Tech  
Email: [wenyou@virginia.edu](mailto:wenyou@virginia.edu)